PHASE 7 OF 8

Project Documentation

Recording, explaining and supporting the LegalEase system

Project: LegalEase – AI-Powered Legal Document Generator

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7.1 Purpose of this Phase

The Project Documentation phase records everything needed to understand, install, use and maintain LegalEase. Good documentation lets reviewers see what was built and why, and lets new users or team members run the application without help from the original developers.

7.2 Documentation Contents

The project documentation records the following:

Section	What it records
Project description	What LegalEase is and the problem it addresses
Objectives	Goals of generating, previewing, editing and exporting drafts
Requirements	Functional and technical requirements
Architecture	Frontend, backend, AI core and formatting modules, with the data flow
Technologies	Python, Streamlit, FastAPI, Uvicorn, Gemini, python-docx, FPDF, Pillow, Requests, python-dotenv
Development activities	Backend, frontend and document-generation work
Testing procedure	Test cases and how they are executed
Setup instructions	How to install and start the application
Application workflow	Step-by-step user journey

7.3 Setup Guide

1. Install Python and create a virtual environment.
2. Install the required packages (Streamlit, FastAPI, Uvicorn, Google Gemini SDK, python-docx, FPDF, Pillow, Requests, python-dotenv).
3. Add the Gemini API key to the .env file.
4. Start the FastAPI backend with Uvicorn.
5. Start the Streamlit frontend.

6. Open the displayed local address in a web browser.

7.4 User Guide: Application Workflow

1. Choose the document type.
2. Enter the parties involved.
3. Enter the terms and conditions.
4. Enter the effective date and jurisdiction.
5. Click *Generate Document* and wait for the draft.
6. Read the preview and edit any part of the text.
7. Download the final document as TXT, DOCX or PDF.

7.5 Document Generation, Editing and Export Features

- **Generation:** Gemini creates structured content from the user's inputs.
- **Editing:** the preview is editable, and exports use the edited text.
- **TXT:** plain text for maximum compatibility.
- **DOCX:** formatted Word file created with python-docx, suitable for further editing.
- **PDF:** formatted PDF created with FPDF, suitable for sharing and printing.
- **Branding:** headings, logo and footer information can appear in formatted outputs.

7.6 Important Disclaimer

The generated documents are drafts and should be reviewed appropriately before actual legal use. LegalEase supports drafting; it does not provide legal advice or guarantee legal validity.

7.7 Troubleshooting

Problem	Likely cause	Suggested action
No document is generated	Backend not running or wrong address	Start Uvicorn and check the backend address
Error from the AI service	Missing or invalid API key, or no internet	Check the .env file and connectivity
Validation error	A required field is empty or wrongly formatted	Complete all inputs and retry
Export file looks unformatted	Wrong export option chosen	Use DOCX or PDF instead of TXT

7.8 Maintenance and Future Updates

Because the frontend, backend, AI core and formatters are separate modules, each can be improved independently. Documentation should be updated whenever dependencies, document types or export options change, so that it always matches the running application.

7.9 Documentation Standards

- Use clear, simple language that a non-technical reader can follow.
- Keep the documents consistent with the running application.
- Describe what each feature does and how to use it, not only how it was built.
- Include the legal-draft disclaimer wherever generated documents are described.

7.10 Technology Reference

Technology	Role
Streamlit	User interface
FastAPI and Uvicorn	Backend API and server
Google Gemini	Content generation
python-docx	DOCX creation
FPDF	PDF creation
Pillow	Image handling
Requests	Frontend-to-backend calls
python-dotenv	Environment configuration

7.11 Intended Audience of the Documents

The documentation serves three audiences: end users who need the workflow and disclaimer, evaluators who need the objectives, architecture and testing evidence, and future developers who need the setup steps and module descriptions.

7.12 Documentation Review Checklist

Check	Done when
Accuracy	Steps and feature descriptions match the running application
Completeness	Description, requirements, architecture, testing and setup are all present
Clarity	A new reader can install and run LegalEase using the guide alone

Check	Done when
Disclaimer	Draft-review notice is included
Team credit	All four team members are listed

7.13 Version and Change Record

Any later change to the project, such as a new document type, a new export format or a different setup step, should be added to the documentation with a short note describing what changed and why, so the record stays trustworthy.

7.14 Summary

With the documentation in place, LegalEase can be understood, installed, used and extended by someone who was not part of the original development, which completes the written record of the project.